1 Equations in ρ coordinates

$$0 = \frac{1}{V'}\frac{\partial}{\partial\rho}\left[V'\langle|\nabla\rho|^2\rangle\frac{\partial}{\partial\rho}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_\mathrm{e} + Z_\mathrm{i}en_\mathrm{i} + Z_\mathrm{z}en_\mathrm{z} + Z_\mathrm{b}en_\mathrm{b} + Z_\mathrm{b}egn_\mathrm{b}^\mathrm{rp}\right) \tag{1.1}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi}_t = {} & V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\left[\frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\psi_t\right] \\ & + \frac{\mu_0}{\langle B^\theta\rangle}\left[-en_\mathrm{e}\left(\langle Bu_{\mathrm{e}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle\right) + Z_\mathrm{i}en_\mathrm{i}\left(\langle Bu_{\mathrm{i}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle\right)\right. \\ & \left. + Z_\mathrm{z}en_\mathrm{z}\left(\langle Bu_{\mathrm{z}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right) + Z_\mathrm{b}en_\mathrm{b}\left(\langle Bu_{\mathrm{b}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{1.2}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi} = {} & \frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\left[V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\psi\right] \\ & - \frac{\mu_0}{\langle R^{-2}\rangle}\left(-en_\mathrm{e}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle + Z_\mathrm{i}en_\mathrm{i}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle + Z_\mathrm{z}en_\mathrm{z}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + Z_\mathrm{b}en_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\end{aligned} \tag{1.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{1.4}$$

$$\frac{\partial}{\partial t}\psi_t = \dot{\psi}_t \tag{1.5}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V'n_\mathrm{e}) = {} & -\frac{1}{V'}\frac{\partial}{\partial\rho}[V'n_\mathrm{e}(u_\mathrm{e}^\rho - u_g^\rho)] + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\ & + Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e} - n_\mathrm{esup}) + (1 - f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{1.6}$$

$$\begin{aligned}0 = {} & -(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}n_\mathrm{e}T_\mathrm{e} + en_\mathrm{e}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}\Phi \\ & - e\frac{\partial\psi}{\partial\rho}In_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + e\frac{\partial\psi}{\partial\rho}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle\end{aligned} \tag{1.7}$$

$$\begin{aligned}m_\mathrm{e}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\parallel}\rangle = {} & -\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}} + \hat{\mu}_2^\mathrm{e}BV_{2\mathrm{e}} \\ & + \ell_{11}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\parallel}\rangle \\ & - \ell_{12}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\ & - en_\mathrm{e}\langle BE_\parallel\rangle - \nu_\mathrm{L}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \nu_{0\mathrm{e}}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle \\ & - \frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\\
&-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle-\frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\\
&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right] \qquad (1.8)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V'm_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle)=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle+(u_{\mathrm{e}}^{\rho}-u_g^{\rho})m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\langle Ru_{\mathrm{e}\zeta}\rangle+\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\right]\\
&+\ell_{11}^{\mathrm{ee}}\langle Ru_{\mathrm{e}\zeta}\rangle+\ell_{11}^{\mathrm{ei}}\langle Ru_{\mathrm{i}\zeta}\rangle+\ell_{11}^{\mathrm{ez}}\langle Ru_{\mathrm{z}\zeta}\rangle+\ell_{11}^{\mathrm{eb}}\langle Ru_{\mathrm{b}\zeta}\rangle\\
&-\ell_{12}^{\mathrm{ee}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle-\ell_{12}^{\mathrm{ei}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle-\ell_{12}^{\mathrm{ez}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle\\
&-en_{\mathrm{e}}\langle RE_{\zeta}\rangle-e\frac{\partial\psi}{\partial\rho}n_{\mathrm{e}}u_{\mathrm{e}}^{\rho}\\
&-\frac{e\langle|\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}}\frac{\partial\psi}{\partial\rho}\left[(1-C_{\mathrm{T}})T_{\mathrm{e}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}+C_{\mathrm{T}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}+n_{\mathrm{e}}T_{\mathrm{e}}\frac{\partial\psi}{\partial\rho}C_{\mathrm{p}}\right]\\
&-\nu_{\mathrm{L}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle-\nu_{0\mathrm{e}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle \qquad (1.9)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{1}{V'^{5/3}}\frac{\partial}{\partial t}(V'^{5/3}n_{\mathrm{e}}T_{\mathrm{e}})=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[-\chi_{\mathrm{e}}\langle|\nabla\rho|^2\rangle n_{\mathrm{e}}\frac{\partial}{\partial\rho}T_{\mathrm{e}}+V_{\mathrm{e}}^{\mathrm{hp}}\langle|\nabla\rho|\rangle n_{\mathrm{e}}T_{\mathrm{e}}+\frac{5}{2}n_{\mathrm{e}}T_{\mathrm{e}}(u_{\mathrm{e}}^{\rho}-u_g^{\rho})\right]\\
&-u_g^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}-u_{\mathrm{i}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{i}}T_{\mathrm{i}}-u_{\mathrm{z}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{z}}T_{\mathrm{z}}-\hat{u}_{\mathrm{i}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{i}}\rangle-\hat{u}_{\mathrm{z}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle\\
&-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{i}}}{\tau_{\mathrm{ei}}}-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{z}}}{\tau_{\mathrm{ez}}}\\
&-\dot{\psi}_t\langle B^{\theta}\rangle(-en_{\mathrm{e}}\hat{u}_{\mathrm{e}\theta}+Z_{\mathrm{i}}en_{\mathrm{i}}\hat{u}_{\mathrm{i}\theta}+Z_{\mathrm{z}}en_{\mathrm{z}}\hat{u}_{\mathrm{z}\theta})\\
&+\dot{\psi}\left(-en_{\mathrm{e}}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle+Z_{\mathrm{i}}en_{\mathrm{i}}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle+Z_{\mathrm{z}}en_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right)\\
&-\nu_{\mathrm{L}}T_{\mathrm{e}}(n_{\mathrm{e}}-n_{\mathrm{esup}})-\frac{3}{2}\nu_{\mathrm{L}T_{\mathrm{e}}}n_{\mathrm{e}}(T_{\mathrm{e}}-T_{\mathrm{esup}})\\
&-E_{\mathrm{H}}\nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\
&+f_{\mathrm{ei}}P_{\mathrm{b}}+P_{\alpha\mathrm{e}}+P_{\mathrm{RFe}}-P_{\mathrm{Br}} \qquad (1.10)
\end{aligned}
$$

$$
\frac{5}{2}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle=-\hat{\mu}_2^{\mathrm{e}}\langle Bu_{\mathrm{e}\parallel}\rangle-\hat{\mu}_3^{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle+\hat{\mu}_2^{\mathrm{e}}BV_{1\mathrm{e}}+\hat{\mu}_3^{\mathrm{e}}BV_{2\mathrm{e}}
$$

$$
\begin{aligned}
&- \ell_{21}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\|}\rangle - \ell_{21}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\|}\rangle - \ell_{21}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\|}\rangle - \ell_{21}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\|}\rangle \\
&+ \ell_{22}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \ell_{22}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \ell_{22}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle
\end{aligned} \tag{1.11}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\mathrm{i}}) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V' n_{\mathrm{i}}(u_{\mathrm{i}}^{\rho} - u_g^{\rho})] + \nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\
&- \nu_{\mathrm{L}}(n_{\mathrm{i}} - n_{\mathrm{isup}}) + \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}}\frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}
\end{aligned} \tag{1.12}
$$

$$
\begin{aligned}
0 = &-(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} - Z_{\mathrm{i}} e n_{\mathrm{i}}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho}\Phi \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho} I\langle Bu_{\mathrm{i}\|}\rangle - Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho}\langle B^2\rangle\langle Ru_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.13}
$$

$$
\begin{aligned}
m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial t}\langle Bu_{\mathrm{i}\|}\rangle = &-\hat{\mu}_1^{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \hat{\mu}_2^{\mathrm{i}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\mu}_1^{\mathrm{i}} BV_{1\mathrm{i}} + \hat{\mu}_2^{\mathrm{i}} BV_{2\mathrm{i}} \\
&+ \ell_{11}^{\mathrm{ie}}\langle Bu_{\mathrm{e}\|}\rangle + \ell_{11}^{\mathrm{ii}}\langle Bu_{\mathrm{i}\|}\rangle + \ell_{11}^{\mathrm{iz}}\langle Bu_{\mathrm{z}\|}\rangle + \ell_{11}^{\mathrm{ib}}\langle Bu_{\mathrm{b}\|}\rangle \\
&- \ell_{12}^{\mathrm{ie}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \ell_{12}^{\mathrm{ii}}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \ell_{12}^{\mathrm{iz}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle BE_{\|}\rangle - \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}}\langle Bu_{\mathrm{b}\|}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&+ \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle - \frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right]
\end{aligned} \tag{1.14}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle) = &-\frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle + (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\langle Ru_{\mathrm{i}\zeta}\rangle + \langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle\right] \\
&+ \ell_{11}^{\mathrm{ie}}\langle Ru_{\mathrm{e}\zeta}\rangle + \ell_{11}^{\mathrm{ii}}\langle Ru_{\mathrm{i}\zeta}\rangle + \ell_{11}^{\mathrm{iz}}\langle Ru_{\mathrm{z}\zeta}\rangle + \ell_{11}^{\mathrm{ib}}\langle Ru_{\mathrm{b}\zeta}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&- \ell_{12}^{\mathrm{ie}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle - \ell_{12}^{\mathrm{ii}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle - \ell_{12}^{\mathrm{iz}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle RE_{\zeta}\rangle + Z_{\mathrm{i}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{i}} u_{\mathrm{i}}^{\rho} \\
&+ Z_{\mathrm{i}} \frac{e\langle |\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left[(1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle - \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle \\
&- \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.15}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{i}} T_{\mathrm{i}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[-\chi_{\mathrm{i}} \langle |\nabla\rho|^2\rangle n_{\mathrm{i}} \frac{\partial}{\partial \rho} T_{\mathrm{i}} + V_{\mathrm{i}}^{\mathrm{hp}} \langle |\nabla\rho|\rangle n_{\mathrm{i}} T_{\mathrm{i}} + \frac{5}{2} n_{\mathrm{i}} T_{\mathrm{i}} (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{e}}}{\tau_{\mathrm{ie}}} - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{z}}}{\tau_{\mathrm{iz}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{i}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{i}}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \langle B u_{\mathrm{b}\parallel}\rangle \frac{\langle B u_{\mathrm{i}\parallel}\rangle}{\langle B^2\rangle} + P_{\alpha\mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{1.16}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{i}} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle = &- \hat{\mu}_2^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel}\rangle - \hat{\mu}_3^{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \hat{\mu}_2^{\mathrm{i}} B V_{1\mathrm{i}} + \hat{\mu}_3^{\mathrm{i}} B V_{2\mathrm{i}} \\
&- \ell_{21}^{\mathrm{ie}} \langle B u_{\mathrm{e}\parallel}\rangle - \ell_{21}^{\mathrm{ii}} \langle B u_{\mathrm{i}\parallel}\rangle - \ell_{21}^{\mathrm{iz}} \langle B u_{\mathrm{z}\parallel}\rangle - \ell_{21}^{\mathrm{ib}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&+ \ell_{22}^{\mathrm{ie}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \ell_{22}^{\mathrm{ii}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \ell_{22}^{\mathrm{iz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle
\end{aligned} \tag{1.17}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} [V' n_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho})] + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 \\
&- \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}}
\end{aligned} \tag{1.18}
$$

$$
\begin{aligned}
0 = &- (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} - Z_{\mathrm{z}} e n_{\mathrm{z}} (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} \Phi \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{z}\parallel}\rangle - Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} \langle B^2\rangle \langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{1.19}
$$

$$
\begin{aligned}
m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel}\rangle = &- \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel}\rangle - \hat{\mu}_2^{\mathrm{z}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle + \hat{\mu}_1^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_2^{\mathrm{z}} B V_{2\mathrm{z}} \\
&+ \ell_{11}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{zi}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{zz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle B E_{\parallel} \rangle - \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&+ \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{\langle B^2 \rangle}{I} \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right] \qquad (1.20)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \langle R u_{\mathrm{z}\zeta} \rangle + \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \right] \\
&+ \ell_{11}^{\mathrm{ze}} \langle R u_{\mathrm{e}\zeta} \rangle + \ell_{11}^{\mathrm{zi}} \langle R u_{\mathrm{i}\zeta} \rangle + \ell_{11}^{\mathrm{zz}} \langle R u_{\mathrm{z}\zeta} \rangle + \ell_{11}^{\mathrm{zb}} \langle R u_{\mathrm{b}\zeta} \rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle R \hat{q}_{\mathrm{e}\zeta} \rangle - \ell_{12}^{\mathrm{zi}} \langle R \hat{q}_{\mathrm{i}\zeta} \rangle - \ell_{12}^{\mathrm{zz}} \langle R \hat{q}_{\mathrm{z}\zeta} \rangle \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle R E_{\zeta} \rangle + Z_{\mathrm{z}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{z}} u_{\mathrm{z}}^{\rho} \\
&+ Z_{\mathrm{z}} \frac{e \langle |\nabla \rho|^2 \rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} \left[(1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \qquad (1.21)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{z}} T_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[- \chi_{\mathrm{z}} \langle |\nabla \rho|^2 \rangle n_{\mathrm{z}} \frac{\partial}{\partial \rho} T_{\mathrm{z}} + V_{\mathrm{z}}^{\mathrm{hp}} \langle |\nabla \rho| \rangle n_{\mathrm{z}} T_{\mathrm{z}} + \frac{5}{2} n_{\mathrm{z}} T_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{e}}}{\tau_{\mathrm{ze}}} - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{i}}}{\tau_{\mathrm{zi}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{z}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{z}}} n_{\mathrm{z}} (T_{\mathrm{z}} - T_{\mathrm{zsup}}) \\
&+ P_{\alpha \mathrm{z}} + P_{\mathrm{RFz}} \qquad (1.22)
\end{aligned}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle = &- \hat{\mu}_2^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\mu}_3^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_3^{\mathrm{z}} B V_{2\mathrm{z}} \\
&- \ell_{21}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel} \rangle - \ell_{21}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel} \rangle - \ell_{21}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel} \rangle - \ell_{21}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \ell_{22}^{\mathrm{ze}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \ell_{22}^{\mathrm{zi}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \ell_{22}^{\mathrm{zz}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle
\end{aligned}
$$

$$- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\|} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\|} \rangle \tag{1.23}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\mathrm{b}}) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V' n_{\mathrm{b}}(u_{\mathrm{b}}^{\rho} - u_g^{\rho})] + \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} - g\frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} - \nu_{\mathrm{b}} n_{\mathrm{b}} - \nu_{\mathrm{bL}} n_{\mathrm{b}} - \nu_{\mathrm{eff}} g(\sqrt{\delta} n_{\mathrm{b}} - n_{\mathrm{b}}^{\mathrm{rp}}) \\ &+ \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle f_t D_{\mathrm{b}} \frac{\partial}{\partial \rho} n_{\mathrm{b}}\end{aligned} \tag{1.24}$$

$$\begin{aligned}0 = &-(\langle B^2 \rangle \langle R^2 \rangle - I^2)\frac{\partial}{\partial \rho} n_{\mathrm{b}} T_{\mathrm{b}} - Z_{\mathrm{b}} e n_{\mathrm{b}} (\langle B^2 \rangle \langle R^2 \rangle - I^2)\frac{\partial}{\partial \rho}\Phi \\ &+ Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{b}\|} \rangle - Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} \langle B^2 \rangle \langle R u_{\mathrm{b}\zeta} \rangle\end{aligned} \tag{1.25}$$

$$\begin{aligned}m_{\mathrm{b}} \frac{\partial}{\partial t} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle = &-(\hat{\mu}_1^{\mathrm{b}} - \ell_{11}^{\mathrm{bb}}) \langle B u_{\mathrm{b}\|} \rangle + \sum_{a \neq \mathrm{b}} \ell_{11}^{\mathrm{b}a} \langle B u_{a\|} \rangle + \hat{\mu}_1^{\mathrm{b}} B V_{1\mathrm{b}} + Z_{\mathrm{b}} e n_{\mathrm{b}} \langle B E_{\|}^{A} \rangle \\ &- \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{0\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + \langle B S_{\mathrm{m}\|\mathrm{b}} \rangle \\ &- \nu_{\mathrm{eff}} g \sqrt{\delta} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle m_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle f_t D_{\mathrm{b}} \frac{\partial}{\partial \rho} n_{\mathrm{b}}\end{aligned} \tag{1.26}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V'(u_{\mathrm{b}}^{\rho} - u_g^{\rho}) m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle] \\ &+ \ell_{11}^{\mathrm{be}} \langle R u_{\mathrm{e}\zeta} \rangle + \ell_{11}^{\mathrm{bi}} \langle R u_{\mathrm{i}\zeta} \rangle + \ell_{11}^{\mathrm{bz}} \langle R u_{\mathrm{z}\zeta} \rangle + \ell_{11}^{\mathrm{bb}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle \\ &+ Z_{\mathrm{b}} e n_{\mathrm{b}} \dot{\psi} + Z_{\mathrm{b}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{b}} u_{\mathrm{b}}^{\rho} - \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle \\ &- \nu_{0\mathrm{i}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle + \frac{I}{\langle B^2 \rangle} \langle B S_{\mathrm{m}\|\mathrm{b}} \rangle\end{aligned} \tag{1.27}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{01}) = &\frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{01} \frac{\partial}{\partial \rho} n_{01} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{01} \\ &- \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} + \gamma_0 \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_{\mathrm{i}}^{\mathrm{puff}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}}\end{aligned} \tag{1.28}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{02}) = \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{02} \frac{\partial}{\partial \rho} n_{02} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{02} + \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} \tag{1.29}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{03}) = \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{03} \frac{\partial}{\partial \rho} n_{03} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{03} + f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \tag{1.30}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' \bar{n}_z) = &\frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{nz} \frac{\partial}{\partial \rho} \bar{n}_z + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \Delta D_{\mathrm{C0}} \frac{\partial}{\partial \rho} n_{\mathrm{C0}} - n_{\mathrm{e}} \bar{n}_z f_5 S_5 + n_{\mathrm{e}} n_{\mathrm{z}} A_6 \\ &+ \gamma_{0\mathrm{z}} \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + \gamma_{\mathrm{i,z}} \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) - (1 - \gamma_{0\mathrm{z}}) \nu_{\mathrm{L}} (\bar{n}_z - n_{\mathrm{C0}}) \\ &+ \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_{\mathrm{z}}^{\mathrm{puff}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}}\end{aligned} \tag{1.31}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\rm b}^{\rm rp}) = \frac{P_{\rm b\perp}}{E_{\rm b}} + \nu_{\rm eff}(\sqrt{\delta} n_{\rm b} - n_{\rm b}^{\rm rp}) - \nu_{\rm b} n_{\rm b}^{\rm rp} - \frac{1}{V'}\frac{\partial}{\partial \rho} V' \left(\langle |\nabla \rho| \rangle u_{\rm b}^{\rm rp} n_{\rm b}^{\rm rp} - \langle |\nabla \rho|^2 \rangle D_{\rm b}^{\rm rp} \frac{\partial}{\partial \rho} n_{\rm b}^{\rm rp} \right) \quad (1.32)$$

2 Additional important relationship

$$\bar{n}_z = \sum_{i=0}^{\max(Z)-1} n_i \quad (2.1)$$

$$B_p = \frac{|\nabla \rho|}{R}\frac{\partial \psi}{\partial \rho}, \quad \langle B_p^2 \rangle = \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.2)$$

$$B_t = \frac{I}{R} = \frac{1}{R}\frac{4\pi^2}{\langle R^{-2} \rangle}\frac{\partial \psi_t}{\partial V}, \quad \langle B_t^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle}\left(\frac{\partial \psi_t}{\partial V} \right)^2 \quad (2.3)$$

$$\langle B^2 \rangle = \langle B_t^2 \rangle + \langle B_p^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle}\left(\frac{\partial \psi_t}{\partial V} \right)^2 + \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.4)$$

$$\langle B^\theta \rangle = \frac{4\pi^2}{V'}\frac{\partial \psi}{\partial \rho} = 4\pi^2 \frac{\partial \psi}{\partial V} = \frac{I \langle R^{-2} \rangle}{q} = 2\pi \left(\oint \frac{dl_p}{B_p} \right)^{-1} \quad (2.5)$$

$$\hat{q}^2 \equiv \frac{I^2}{2\langle B_p^2 \rangle}\left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \quad (2.6)$$

$$\langle B^2 \rangle \langle R^2 \rangle - I^2 = \langle R^2 \rangle \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 (1 + 2\hat{q}^2) \quad (2.7)$$

$$\langle B \hat{q}_{s\|} \rangle \equiv \frac{2\langle B q_{s\|} \rangle}{5 p_s} \quad (2.8)$$

$$\hat{u}_{s\theta} = \frac{\langle B u_{s\|} \rangle - B V_{1s}}{\langle B^2 \rangle} \quad (2.9)$$

$$\begin{aligned} \left\langle \frac{u_{s\zeta}}{R} \right\rangle &= \frac{1}{1+2\hat{q}^2}\frac{1}{\langle R^2 \rangle}\langle R u_{s\zeta} \rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I}\langle B u_{s\|} \rangle \quad (2.10) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + \frac{I}{\langle B^2 \rangle}\left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right)(\langle B u_{s\|} \rangle - B V_{1s}) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + I \left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \hat{u}_{s\theta} \end{aligned}$$

$$\langle R \hat{q}_{s\zeta} \rangle = \frac{1}{\langle B^2 \rangle}\left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right)\frac{1}{I} B V_{2s} + \frac{I}{\langle B^2 \rangle}\langle B \hat{q}_{s\|} \rangle \quad (2.11)$$

$$B V_{1s} = -I \left(\frac{1}{e_s n_s}\frac{\partial n_s T_s}{\partial \psi} + \frac{\partial \Phi}{\partial \psi} \right), \quad B V_{2s} = -\frac{I}{e_s}\frac{\partial T_s}{\partial \psi} \quad (2.12)$$

$$u^\rho \equiv \langle \boldsymbol{u}\cdot\nabla\rho\rangle, \quad u^V = \langle \boldsymbol{u}\cdot\nabla V\rangle \tag{2.13}$$

$$u^\rho_g = -\frac{4\pi^2}{IV'\langle R^{-2}\rangle}\dot{\psi}_t, \quad u^V_g = -\frac{4\pi^2}{I\langle R^{-2}\rangle}\dot{\psi}_t = -\frac{V'}{2\rho\psi_{ta}}\dot{\psi}_t \tag{2.14}$$

$$\frac{1}{V'}\frac{\partial V'}{\partial t} = \frac{\partial}{\partial V}\langle \boldsymbol{u}_g\cdot\nabla V\rangle = \frac{\partial u^V_g}{\partial V} \tag{2.15}$$

$$\langle E_r\rangle = -\langle|\nabla\rho|^2\rangle\frac{\partial r}{\partial\rho}\frac{\partial}{\partial\rho}\Phi \tag{2.16}$$

$$\langle BE_\parallel\rangle = I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) \tag{2.17}$$

$$RE_\zeta = \dot{\psi} = \frac{V_{\rm loop}}{2\pi} \tag{2.18}$$

$$\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\rm i}\rangle = \hat{\mu}^{\rm i}_1(\langle Bu_{\rm i\parallel}\rangle - BV_{1\rm i}) + \hat{\mu}^{\rm i}_2\left(\frac{2\langle Bq_{\rm i\parallel}\rangle}{5p_{\rm i}} - BV_{2\rm i}\right) \tag{2.19}$$

$$\hat{\ell}^{ab}_{ij} \equiv \frac{\ell^{ab}_{ij}}{m_a n_a}, \quad \hat{\ell}^{\rm bb}_{ij} \equiv \frac{\ell^{\rm bb}_{ij}}{m_{\rm e}n_{\rm e}} \tag{2.20}$$

$$\Gamma^{\rm puff} \simeq \frac{\langle \boldsymbol{\Gamma}^{\rm puff}\cdot\nabla\rho\rangle}{\langle|\nabla\rho|\rangle} \tag{2.21}$$

3 Equations in the coordinates suitable for implementation

$$0 = \frac{\partial}{\partial V}\left[\langle|\nabla V|^2\rangle\frac{\partial}{\partial V}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_{\rm e} + Z_{\rm i}en_{\rm i} + Z_{\rm z}en_{\rm z} + Z_{\rm b}en_{\rm b} + Z_{\rm b}eg n^{\rm rp}_{\rm b}\right) \tag{3.1}$$

$$\begin{aligned}\epsilon_0\frac{\partial}{\partial t}\dot{\psi}_t = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\right] - \frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\right]\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\\ & + \frac{1}{\langle B^\theta\rangle}\left[-e\left(n_{\rm e}\langle Bu_{\rm e\parallel}\rangle - In_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle\right) + Z_{\rm i}e\left(n_{\rm i}\langle Bu_{\rm i\parallel}\rangle - In_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle\right)\right.\\ & \left. + Z_{\rm z}e\left(n_{\rm z}\langle Bu_{\rm z\parallel}\rangle - In_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle\right) + Z_{\rm b}e\left(n_{\rm b}\langle Bu_{\rm b\parallel}\rangle - In_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{3.2}$$

$$\begin{aligned}\frac{1}{c^2}\left\langle\frac{1}{R^2}\right\rangle\frac{\partial}{\partial t}\dot{\psi} = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial}{\partial V}\psi\right]\\ & - \mu_0\left(-en_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle + Z_{\rm i}en_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle + Z_{\rm z}en_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle + Z_{\rm b}en_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\end{aligned} \tag{3.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{3.4}$$

$$\frac{\partial}{\partial t}\frac{\psi_t}{\mu_0} = \frac{\dot{\psi}_t}{\mu_0} \tag{3.5}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e} = &-\frac{\partial}{\partial V}n_\mathrm{e}u_\mathrm{e}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{e} + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\ &+ Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e}-n_\mathrm{esup}) + (1-f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{3.6}$$

$$0 = -\frac{e}{m_\mathrm{e}}n_\mathrm{e}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}BV_{1\mathrm{e}} - \frac{e}{m_\mathrm{e}}In_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \frac{e}{m_\mathrm{e}}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle \tag{3.7}$$

$$\begin{aligned}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\|}\rangle = &-\frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}}\\ &+ \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{1}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}T_\mathrm{e} - \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{T_\mathrm{e}}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}n_\mathrm{e}\langle Bu_{\mathrm{i}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}n_\mathrm{e}\langle Bu_{\mathrm{z}\|}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle\\ &- \hat{\ell}_{12}^{\mathrm{ee}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ei}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ez}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{z}\|}\rangle\\ &- e\frac{n_\mathrm{e}}{m_\mathrm{e}}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) - \nu_\mathrm{L}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \frac{\partial}{\partial V}\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &- \frac{\partial}{\partial V}\left[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right.\\ &\left.-\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\partial}{\partial V}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right]\end{aligned} \tag{3.8}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle = &-(u_\mathrm{e}^V-u_g^V)\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial u_\mathrm{e}^V}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle]\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\langle Ru_{\mathrm{e}\zeta}\rangle\frac{\partial}{\partial V}n_\mathrm{e} - \frac{1}{m_\mathrm{e}}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}\frac{n_\mathrm{e}}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}\frac{n_\mathrm{e}}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}_{12}^{\mathrm{ee}} n_{\mathrm{e}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ee}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ei}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{ei}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- e \frac{n_{\mathrm{e}}}{m_{\mathrm{e}}} \dot{\psi} - e \frac{1}{m_{\mathrm{e}}} \frac{\partial \psi}{\partial V} n_{\mathrm{e}} u_{\mathrm{e}}^V \\
&- \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{e}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[n_{\mathrm{e}} \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{e}} \frac{\partial}{\partial V} \Phi + \frac{C_{\mathrm{T}} - 1}{e} \left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{e}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle
\end{aligned} \tag{3.9}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{e}} T_{\mathrm{e}} = &- \frac{5}{2} (u_{\mathrm{e}}^V - u_g^V) \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{5}{2} \frac{\partial u_{\mathrm{e}}^V}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} [V_{\mathrm{e}}^{\mathrm{hp}} \langle |\nabla V| \rangle n_{\mathrm{e}} T_{\mathrm{e}}] \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- u_g^V \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - u_{\mathrm{i}}^V \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - u_{\mathrm{z}}^V \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle \\
&- \frac{3}{2} \nu_{\mathrm{Tei}} \left(n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{Tez}} \left(n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- 4\pi^2 \frac{\partial \psi}{\partial V} \dot{\psi}_t (-e n_{\mathrm{e}} \hat{u}_{\mathrm{e}\theta} + Z_{\mathrm{i}} e n_{\mathrm{i}} \hat{u}_{\mathrm{i}\theta} + Z_{\mathrm{z}} e n_{\mathrm{z}} \hat{u}_{\mathrm{z}\theta}) \\
&+ \dot{\psi} \left(-e n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + Z_{\mathrm{i}} e n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + Z_{\mathrm{z}} e n_{\mathrm{z}} \left\langle \frac{u_{\mathrm{z}\zeta}}{R} \right\rangle \right) \\
&- \nu_{\mathrm{L}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{esup}} T_{\mathrm{e}}) - \frac{3}{2} \nu_{\mathrm{LT_e}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} T_{\mathrm{esup}}) - E_{\mathrm{H}} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\
&+ f_{\mathrm{ei}} P_{\mathrm{b}} + P_{\alpha \mathrm{e}} + P_{\mathrm{RFe}} - P_{\mathrm{Br}}
\end{aligned} \tag{3.10}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{e}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle = &- \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \frac{1}{m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} B V_{1\mathrm{e}} \\
&+ \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{e}}}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- \hat{\ell}_{21}^{\mathrm{ee}} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ei}} n_{\mathrm{e}} \langle B u_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ez}} n_{\mathrm{e}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{eb}} \frac{n_{\mathrm{e}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ee}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ei}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ez}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned} \tag{3.11}
$$

$$
0 = - n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle \tag{3.12}
$$

$$0 = -n_{\mathrm{e}} B V_{1\mathrm{e}} + \frac{1}{e}\frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.13}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} = &-\frac{\partial}{\partial V} n_{\mathrm{i}} u_{\mathrm{i}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{i}} + \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) - \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) \\ &+ \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}\end{aligned} \tag{3.14}$$

$$0 = Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1\mathrm{i}} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} I n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \langle B^2 \rangle n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \tag{3.15}$$

$$\begin{aligned}n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B u_{\mathrm{i}\parallel} \rangle = &-\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} B V_{1\mathrm{i}} \\ &- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} n_{\mathrm{i}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} n_{\mathrm{i}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\ &- \hat{\ell}_{12}^{\mathrm{ie}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{ii}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{iz}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\ &+ Z_{\mathrm{i}} e \frac{n_{\mathrm{i}}}{m_{\mathrm{i}}} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\ &- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{LC}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &- \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right. \\ &\left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right]\end{aligned} \tag{3.16}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle = &-(u_{\mathrm{i}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial u_{\mathrm{i}}^{V}}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \langle R u_{\mathrm{i}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} - \frac{1}{m_{\mathrm{i}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} \frac{n_{\mathrm{i}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\|} \rangle \\
&+ Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \dot{\psi} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \frac{\partial \psi}{\partial V} n_{\mathrm{i}} u^V_{\mathrm{i}} \\
&+ Z_{\mathrm{i}} \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{i}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[(n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + (n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial}{\partial V} \Phi \right. \\
&\left. + \frac{C_{\mathrm{T}} - 1}{e} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{i}} + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{b}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle
\end{aligned} \tag{3.17}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{i}} T_{\mathrm{i}} = &- \frac{5}{2} (u^V_{\mathrm{i}} - u^V_g) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{5}{2} \frac{\partial u^V_{\mathrm{i}}}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} V^{\mathrm{hp}}_{\mathrm{i}} \langle |\nabla V| \rangle n_{\mathrm{i}} T_{\mathrm{i}} \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle T_{\mathrm{i}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&+ (u^V_{\mathrm{i}} - u^V_g) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} \nu_{\mathrm{Tei}} \left(\frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{e}} T_{\mathrm{e}} \right) - \frac{3}{2} \nu_{\mathrm{Tiz}} \left(n_{\mathrm{i}} T_{\mathrm{i}} - \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- \nu_{\mathrm{L}} \left(n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{isup}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{LT_i}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{1}{n_{\mathrm{b}}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \frac{\langle B u_{\mathrm{i}\|} \rangle}{\langle B^2 \rangle} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + P_{\alpha \mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{3.18}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{i}\|} \rangle = &- \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_2 \langle B u_{\mathrm{i}\|} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 \langle B \hat{q}_{\mathrm{i}\|} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_2 B V_{1\mathrm{i}} \\
&- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&- \hat{\ell}^{\mathrm{ie}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{e}\|} \rangle - \hat{\ell}^{\mathrm{ii}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{i}\|} \rangle - \hat{\ell}^{\mathrm{iz}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{z}\|} \rangle - \hat{\ell}^{\mathrm{ib}}_{21} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{ie}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\|} \rangle + \hat{\ell}^{\mathrm{ii}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle + \hat{\ell}^{\mathrm{iz}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\|} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle
\end{aligned} \tag{3.19}
$$

$$
0 = - n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{i}} \langle B u_{\mathrm{i}\|} \rangle \tag{3.20}
$$

$$0 = -n_{\mathrm{i}} B V_{1\mathrm{i}} - \frac{1}{Z_{\mathrm{i}} e} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.21}$$

$$\frac{\partial}{\partial t} n_{\mathrm{z}} = -\frac{\partial}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{z}} + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 - \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{isup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}} \tag{3.22}$$

$$0 = Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1\mathrm{z}} + Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} I n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} \langle B^2 \rangle n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \tag{3.23}$$

$$\begin{aligned}
n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel} \rangle = & -\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \frac{1}{m_{\mathrm{z}}} \hat{\mu}_1^{\mathrm{i}} B V_{1\mathrm{z}} \\
& - \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{z}}}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} \\
& + \hat{\ell}_{11}^{\mathrm{ze}} n_{\mathrm{z}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} n_{\mathrm{z}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
& - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
& + Z_{\mathrm{z}} e \frac{n_{\mathrm{z}}}{m_{\mathrm{z}}} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\
& - \nu_{\mathrm{L}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
& - \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
& + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
& - \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{z}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
& - \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{z}} \right. \\
& \left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{z}} \right]
\end{aligned} \tag{3.24}$$

$$\begin{aligned}
\frac{\partial}{\partial t} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle = & -(u_{\mathrm{z}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \\
& + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \langle R u_{\mathrm{z}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{z}} - \frac{1}{m_{\mathrm{z}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
& + \hat{\ell}_{11}^{\mathrm{ze}} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} \frac{n_{\mathrm{z}}}{n_{\mathrm{i}}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle \\
& - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned}$$

$$
\begin{aligned}
&+\hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{i}\|}\rangle \\
&+\hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}}\dot{\psi} + Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}}\frac{\partial \psi}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} \\
&+ Z_{\mathrm{z}}\frac{e\langle|\nabla V|^2\rangle D_{\mathrm{e}}}{m_{\mathrm{z}} T_{\mathrm{e}}}\frac{\partial \psi}{\partial V}\left[n_{\mathrm{z}}\frac{\partial \psi}{\partial V}\frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{z}}\frac{\partial}{\partial V}\Phi\right. \\
&\left.+\frac{C_{\mathrm{T}}-1}{e}\frac{n_{\mathrm{z}}}{n_{\mathrm{e}}}\left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}}\frac{\partial}{\partial V} n_{\mathrm{e}}\right) + \frac{T_{\mathrm{e}}}{e}\frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{z}}\right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{3.25}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{\partial}{\partial t} n_{\mathrm{z}} T_{\mathrm{z}} &= -\frac{5}{2}(u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{5}{2}\frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V} V_{\mathrm{z}}^{\mathrm{hp}}\langle|\nabla V|\rangle n_{\mathrm{z}} T_{\mathrm{z}} \\
&+ \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle T_{\mathrm{z}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&+ (u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta}\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle - \frac{3}{2}\nu_{\mathrm{Tez}}\left(\frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{e}} T_{\mathrm{e}}\right) - \frac{3}{2}\nu_{\mathrm{Tiz}}\left(\frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{i}} T_{\mathrm{i}}\right) \\
&- \nu_{\mathrm{L}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{zsup}} T_{\mathrm{z}}\right) - \frac{3}{2}\nu_{\mathrm{LT_z}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}} T_{\mathrm{zsup}}\right) + P_{\alpha\mathrm{z}} + P_{\mathrm{RFz}}
\end{aligned} \tag{3.26}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{z}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{z}\|}\rangle &= -\frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \frac{1}{m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle + \frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} \\
&- \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{1}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{T_{\mathrm{z}}}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&- \hat{\ell}_{21}^{\mathrm{ze}} n_{\mathrm{z}}\langle B u_{\mathrm{e}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zi}} n_{\mathrm{z}}\langle B u_{\mathrm{i}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zz}} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zb}}\frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}}\langle B u_{\mathrm{b}\|}\rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ze}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zi}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zz}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle
\end{aligned} \tag{3.27}
$$

$$
0 = -n_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle \tag{3.28}
$$

$$
0 = -n_{\mathrm{z}} B V_{1\mathrm{z}} - \frac{1}{Z_{\mathrm{z}} e}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V}\Phi \tag{3.29}
$$

$$\frac{\partial}{\partial t}n_\mathrm{b} = -\frac{\partial}{\partial V}n_\mathrm{b}u_\mathrm{b}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{b} + \frac{P_\mathrm{b}}{E_\mathrm{b}} - g\frac{P_{\mathrm{b}\perp}}{E_\mathrm{b}} - \nu_\mathrm{b}n_\mathrm{b} - \nu_\mathrm{bL}n_\mathrm{b} - \nu_\mathrm{eff}g(\sqrt{\delta}n_\mathrm{b} - n_\mathrm{b}^\mathrm{rp})$$
$$+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle f_t D_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} \tag{3.30}$$

$$0 = Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}n_\mathrm{b}BV_{1\mathrm{b}} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}In_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\langle B^2\rangle n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle \tag{3.31}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle = -\frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}BV_{1\mathrm{b}}$$
$$+ \hat{\ell}_{11}^{\mathrm{bb}}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \sum_{a\neq\mathrm{b}}\hat{\ell}_{11}^{\mathrm{b}a}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}}\langle Bu_{a\|}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q} + \dot{\psi}\right) - \nu_\mathrm{bL}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - \nu_{0\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle$$
$$- \nu_\mathrm{CX}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{1}{m_\mathrm{b}}\langle BS_{\mathrm{m}\|\mathrm{b}}\rangle - \nu_\mathrm{eff}g\sqrt{\delta}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\langle Bu_{\mathrm{b}\|}\rangle f_t D_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} \tag{3.32}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle = u_g^V\frac{\partial}{\partial V}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{\partial}{\partial V}\langle Ru_{\mathrm{b}\zeta}\rangle n_\mathrm{b}u_\mathrm{b}^V$$
$$+ \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{be}}\frac{1}{n_\mathrm{e}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bi}}\frac{1}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bz}}\frac{1}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bb}}\frac{1}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}\dot{\psi} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\frac{\partial\psi}{\partial V}n_\mathrm{b}u_\mathrm{b}^V$$
$$- \nu_\mathrm{bL}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{CX}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\frac{I}{\langle B^2\rangle}\langle BS_{\mathrm{m}\|\mathrm{b}}\rangle \tag{3.33}$$

$$0 = -n_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle \tag{3.34}$$

$$0 = -n_\mathrm{b}BV_{1\mathrm{b}} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}In_\mathrm{b}\frac{\partial}{\partial V}T_\mathrm{b} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}IT_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} - n_\mathrm{b}\frac{\partial V}{\partial\psi}I\frac{\partial}{\partial V}\Phi \tag{3.35}$$

$$\frac{\partial}{\partial t}n_{01} = -\frac{\partial u_g^V}{\partial V}n_{01} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{01}\frac{\partial}{\partial V}n_{01} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{01}$$
$$- \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} + \gamma_0\nu_\mathrm{L}(n_\mathrm{i} - n_\mathrm{isup}) + \frac{\partial}{\partial V}V'\Gamma^\mathrm{puff}\Big|_\mathrm{b} \tag{3.36}$$

$$\frac{\partial}{\partial t}n_{02} = -\frac{\partial u_g^V}{\partial V}n_{02} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{02}\frac{\partial}{\partial V}n_{02} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{02} + \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} \tag{3.37}$$

$$\frac{\partial}{\partial t}n_{03} = -\frac{\partial u_g^V}{\partial V}n_{03} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{03}\frac{\partial}{\partial V}n_{03} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{03} + f_\mathrm{CX}\frac{P_\mathrm{b}}{E_\mathrm{b}} \tag{3.38}$$

$$\frac{\partial}{\partial t}\bar{n}_z = -\frac{\partial u_g^V}{\partial V}\bar{n}_z + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{nz}\frac{\partial}{\partial V}\bar{n}_z + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle \Delta D_{\mathrm{C0}}\frac{\partial}{\partial V}n_{\mathrm{C0}} - n_{\mathrm{e}}\bar{n}_z f_5 S_{\bar{5}} + n_{\mathrm{e}} n_{\mathrm{z}} A_6$$
$$+ \gamma_{0\mathrm{z}}\nu_{\mathrm{L}}\left(n_{\mathrm{z}} - n_{\mathrm{zsup}}\right) + \gamma_{\mathrm{i,z}}\nu_{\mathrm{L}}\left(n_{\mathrm{i}} - n_{\mathrm{isup}}\right) - (1-\gamma_{0\mathrm{z}})\nu_{\mathrm{L}}\left(\bar{n}_z - n_{\mathrm{C0}}\right)$$
$$+ \frac{\partial}{\partial V}V'\Gamma_{\mathrm{z}}^{\mathrm{puff}}\Big|_{\mathrm{b}} \tag{3.39}$$

$$\frac{\partial}{\partial t}n_{\mathrm{b}}^{\mathrm{rp}} = -\frac{\partial u_g^V}{\partial V}n_{\mathrm{b}}^{\mathrm{rp}}\frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} + \nu_{\mathrm{eff}}(\sqrt{\delta}n_{\mathrm{b}} - n_{\mathrm{b}}^{\mathrm{rp}}) - \nu_{\mathrm{b}}n_{\mathrm{b}}^{\mathrm{rp}}$$
$$- \frac{\partial}{\partial V}\langle|\nabla V|\rangle u_{\mathrm{b}}^{\mathrm{rp}}n_{\mathrm{b}}^{\mathrm{rp}} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{\mathrm{b}}^{\mathrm{rp}}\frac{\partial}{\partial V}n_{\mathrm{b}}^{\mathrm{rp}} \tag{3.40}$$

4 Boundary Conditions

num.	name	variable	highest order derivative	center	edge
1	LQm1	Φ	2	N	0
2	LQm2	$\dot{\psi}_t$	0	×	×
3	LQm3	$\dot{\psi}$	0	×	×
4	LQm4	ψ	2	N	$2\pi\mu_0 I_\mathrm{p}$
5	LQm5	ψ_t/μ_0	2	0	$\langle\lvert\nabla V\rvert^2/R^2\rangle\lvert_{V_b} I_\mathrm{vac}/(4\pi^2)$
6	LQe1	n_e	2	N	N†
7	LQe2	$n_\mathrm{e}u_\mathrm{e}^V$	1	0	×
8	LQe3	$\langle Bu_{\mathrm{e}\parallel}\rangle$	2(0)	N(×)	N(×)
9	LQe4	$n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle$	2	N	N
10	LQe5	$n_\mathrm{e}T_\mathrm{e}$	2	N	N
11	LQe6	$\langle B\hat{q}_{\mathrm{e}\parallel}\rangle$	0	×	×
12	LQe7	$n_\mathrm{e}\langle u_{\mathrm{e}\zeta}/R\rangle$	0	×	×
13	LQe8	$BV_{1\mathrm{e}}$	0	×	×
14	LQi1	n_i	2	N	N†
15	LQi2	$n_\mathrm{i}u_\mathrm{i}^V$	1	0	×
16	LQi3	$\langle Bu_{\mathrm{i}\parallel}\rangle$	2(0)	N(×)	N(×)
17	LQi4	$n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle$	2	N	N
18	LQi5	$n_\mathrm{i}T_\mathrm{i}$	2	N	N
19	LQi6	$\langle B\hat{q}_{\mathrm{i}\parallel}\rangle$	0	×	×
20	LQi7	$n_\mathrm{i}\langle u_{\mathrm{i}\zeta}/R\rangle$	0	×	×
21	LQi8	$BV_{1\mathrm{i}}$	0	×	×
22	LQz1	n_z	2	N	N†
23	LQz2	$n_\mathrm{z}u_\mathrm{z}^V$	1	0	×
24	LQz3	$\langle Bu_{\mathrm{z}\parallel}\rangle$	2(0)	N(×)	N(×)
25	LQz4	$n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle$	2	N	N
26	LQz5	$n_\mathrm{z}T_\mathrm{z}$	2	N	N
27	LQz6	$\langle B\hat{q}_{\mathrm{z}\parallel}\rangle$	0	×	×
28	LQz7	$n_\mathrm{z}\langle u_{\mathrm{z}\zeta}/R\rangle$	0	×	×
29	LQz8	$BV_{1\mathrm{z}}$	0	×	×
30	LQb1	n_b	0(2)	×(N)	×(N)
31	LQb2	$n_\mathrm{b}u_\mathrm{b}^V$	1	0	×
32	LQb3	$n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle$	0	×	×
33	LQb4	$n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$	0	×	×
34	LQb7	$n_\mathrm{b}\langle u_{\mathrm{b}\zeta}/R\rangle$	0	×	×
35	LQb8	$BV_{1\mathrm{b}}$	0	×	×
36	LQn1	n_{01}	2	N	$\langle\lvert\nabla V\rvert\rangle S_\mathrm{gas}\rvert_b$
37	LQn2	n_{02}	2	N	0
38	LQn3	n_{03}	2	N	0
39	LQnz	$\bar{n}_z$	2	N	$\langle\lvert\nabla V\rvert\rangle S_{\mathrm{gas,z}}\rvert_b$
40	LQr1	n_b^rp	0(2)	×(N)	×(N)

0: Dirichlet condition ($u = 0$), N: Neumann condition ($u' = 0$)

value: Neumann condition ($u' =$ value), ×: no condition

†: imposed as a constraint in V5.60 (`MDLWBC` = 1), see Sec. 5.2

NOTE: The dependent variable of LQm4 turns to be ψ/μ_0 when NB is applied. Subsequently, the edge boundary condition listed for that variable is also changed to $2\pi I_\mathrm{p}$.

NOTE: Rows 2 to 5 deserve a word, because the variable that carries the second-order operator is not the variable of the equation in which the operator is written. The LQm2 equation advances $\dot{\psi}_t$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \langle R^{-2}\rangle^{-1} \frac{\partial}{\partial V}(\psi_t/\mu_0)]$; the LQm3 equation advances $\dot{\psi}$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \frac{\partial}{\partial V}\psi]$. Counting per dependent variable over the whole system, as this section does, $\dot{\psi}_t$ and $\dot{\psi}$ are never differentiated and so take order 0 and no condition of their own, while ψ_t/μ_0 and ψ take order 2 and carry the two conditions each. The table above now reads that way. In the implementation the two conditions of each pair are applied to the LQm2 and LQm3 equation rows, since that is where the boundary term left by the integration by parts appears; this is the correct weak form and is not a discrepancy with the table. The axis condition of ψ_t/μ_0 is imposed there as $\dot{\psi}_t|_{V=0} = 0$, which holds $\psi_t(0)$ at its initial value, namely zero, and is therefore the Dirichlet condition listed.

NOTE: In the table the column labeled by the highest order derivative means the order of the highest order derivative of a dependent variable, not an equation, in the corresponding row. Please note that the term not integrated by parts is never counted. For example, the first row shows the highest order of derivatives is equal to 2 for Φ because the LQm1 equation has the diffusion term having the second derivative of Φ with respect to V. In addition, the LQe2 and LQi2 equations also have the first derivative of Φ, which is not a dependent variable of these equations. As a whole, the highest order derivative of Φ in this equation system is 2 that comes from the LQm1 equation. When we look at n_e, we find that the highest order derivative of n_e is 2 shown in the LQe5 equation, even though no derivative appears in the LQe1 equation. The Neumann conditions for n_e come from the LQe5 equation. When it comes to $n_\mathrm{e}u_\mathrm{e}^V$, the highest derivative is found as order 1 in the LQe1 equation, but this term is not integrated by parts. It means that any boundary condition is not naturally imposed on $n_\mathrm{e}u_\mathrm{e}^V$. The Dirichlet boundary condition should be imposed at axis.

5 Models, switches and defaults of V5.60

V5.60 changes the default SOL and neutral physics. The detailed physics, the verification and the development history are in `neutral_transport.tex`; this section states what the code solves.

5.1 SOL parallel loss

Outside the separatrix the parallel loss rate is

$$\nu_{\mathrm{L}} = k_L \frac{C_s}{L_c}\, g(\rho), \qquad L_c = 2\pi q R, \qquad g = \frac{x^2}{1+x^2}, \quad x = \frac{r-a}{\delta}\ (r \ge a),\ x = 0\ (r < a), \tag{5.1}$$

and enters every SOL equation in Robin form, e.g. $-\nu_{\mathrm{L}}(n_s - n_{\mathrm{ssup}})$. k_L is `FSLP`. The ramp $g(\rho)$ is numerical smoothing: it keeps ν_{L} from jumping at $\rho = 1$, which the FEM cannot represent, and its continuum limit is the step. `MDLSOL` selects its width δ:

`MDLSOL`	δ
0	$3\sqrt{0.5\,T_i(0)\,m_i}\,q(a)/(Z_i e B_\phi(a))/\sqrt{a/R}$ (V5.52)
1	ion banana width at the separatrix, $\sqrt{\epsilon}\,\sqrt{T_i m_i}/(Z_i e B_\theta)$ (default)

With `MDLSOL` $= 0$ the width is 1.8 times the whole SOL for the reference equilibrium, so g is 0.17 at the wall and the ramp acts as a suppression factor rather than as smoothing. `MDLSOL` $= 1$ gives $g \simeq 0.99$ at the wall. The default $k_L = 1$ is the sheath-limited two-point value when $2\pi qR$ is read as the target-to-target length ($n_{\mathrm{t}} = n_{\mathrm{u}}/2$); V5.52 used the empirical $k_L = 0.3$, which with `MDLSOL` $= 1$ does not give a stable SOL on the X-point equilibrium.

5.2 Wall boundary condition for the densities

The table of the previous section lists a Neumann condition for the densities at the wall $\rho = \rho_b$, said to come naturally from the pressure equation. The natural boundary condition of that equation is, however, zero heat flux, not a condition on the density. Up to V5.52 the densities therefore had no condition at the wall, and the radial particle flux through ρ_b was fixed by the last element: from +57% to −40% of the separatrix flux between `NRMAX` $= 60$ and 120, and −28% even at 480. The radial flux $n_s u_{\mathrm{s}}^V$ cannot take the condition instead: its equation is first order and not integrated by parts, so it already has its one condition at the axis. ρ_b is a computational boundary for the charged species, whose exhaust is carried by ν_{L}.

V5.60 (`MDLWBC` $= 1$) imposes $dn_s/d\rho = 0$ on n_{e}, n_{i} and n_{z}. In the P1 basis this is the constraint

$$n_s(N) = n_s(N-1), \qquad N = \texttt{NRMAX}. \tag{5.2}$$

Constrained degrees of freedom, in general. For a linear system $Ax = b$ with the constraint $x_N = x_{N-1}$, write $x = Ty$, where T is the identity on $y_1 \dots y_{N-1}$ and copies y_{N-1} into x_N. The constrained Galerkin system is

$$T^{\mathrm{T}} A T\, y = T^{\mathrm{T}} b. \tag{5.3}$$

T^{T} adds row N to row $N-1$; AT adds column N to column $N-1$. In finite elements this is the basis reduction $\phi_{N-1} \to \phi_{N-1} + \phi_N$: the trial and test functions are constant on the last element, which is the P1 form of $dn/d\rho = 0$. The code keeps the system square and solves the equivalent

$$\text{row } N-1 \leftarrow \text{row } N-1 + \text{row } N, \qquad \text{row } N \leftarrow x_N - x_{N-1} = 0, \tag{5.4}$$

with x_N still an unknown, so the other equations keep their coupling to it and the column sum is carried by the constraint row. Since $\sum_i \phi_i = 1$, the sum of all rows of a continuity equation is the global particle balance; each column of T^{T} sums to one, so that sum, and with it particle conservation, is unchanged. Replacing row N by the constraint without adding it to row $N-1$ would drop the particle balance of the wall half-cell, an $O(h)$ conservation error.

Implementation. `txcala.f90`, subroutine `BOUNDARY_ZEROGRAD(LQ)`, called from the boundary-condition block of `TXCALA` for `LQe1`, `LQi1` and `LQz1` after the Dirichlet eliminations. In the band storage the coefficient of term `NC` of equation `NQ` in row `NR` is `ALC`, `BLC` or `CLC(NR,NC,NQ)` for `X(NR+1)`, `X(NR)` or `X(NR-1)`, and `PLC` holds the known terms. For every term `NC = 0..NLCMAX`: `BLC(N-1) += CLC(N)`, `ALC(N-1) += BLC(N)`, `PLC(N-1) += PLC(N)`; row `N` is then cleared and set to `BLC(N,1) = 1`, `CLC(N,1) = -1`. The time term and its right-hand side are built from the same coefficient arrays, so they are folded with them. The condition converges at first order: $\Gamma_{\mathrm{wall}}/\Gamma_{\mathrm{sep}}$ is -1.8, -1.1, -0.5 and -0.3% at `NRMAX` = 60, 120, 240 and 480.

5.3 C^0–C^5 superstage (M5)

With `MDLNZD` = 1 the bundle $\bar{n}_z$ is transported as the magnetized C^{1+}–C^{5+} population it mostly is, $D_{nz} = D_{\mathrm{c}}$ (the charged-impurity diffusivity `Dfs`), except for the freshly sputtered C^0. Its lifetime $1/(S_0 n_{\mathrm{e}})$ is microseconds, so its density is slaved to its source:

$$n_{\mathrm{C0}} = \min\left(\frac{S_{\mathrm{wall}}}{S_0 n_{\mathrm{e}}},\ \bar{n}_z\right), \quad S_{\mathrm{wall}} = \gamma_{0\mathrm{z}}\nu_{\mathrm{L}}\max(n_{\mathrm{z}} - n_{\mathrm{zsup}}, 0) + \gamma_{\mathrm{i,z}}\nu_{\mathrm{L}}\max(n_{\mathrm{i}} - n_{\mathrm{isup}}, 0), \tag{5.5}$$

where S_0 is the $C^0 \to C^{1+}$ ionization rate (OpenADAS scd96, `data/Carbon-S0F0_20260921.txt`). n_{C0} enters the $\bar{n}_z$ equation as a known field through the excess flux with $\Delta D_{\mathrm{C0}} = D_{\mathrm{C0}} - D_{\mathrm{c}}$, $D_{\mathrm{C0}} = \bar{v}_{0z}^2/(3S_0 n_{\mathrm{e}})$, and through the parallel loss of the charged part, $(1-\gamma_{0\mathrm{z}})\nu_{\mathrm{L}}(\bar{n}_z - n_{\mathrm{C0}})$ (equations in the previous sections). With `MDLNZD` = 0 (V5.52), $D_{nz} = D_{01}$ and the n_{C0} terms are absent.

Element-based C^0 limiter. D_{C0} is Larsen-limited, $D \to D/(1 + D|\nabla n_{\mathrm{C0}}|/(\alpha \bar{v}_{0z} n_{\mathrm{C0}}))$, α = `FLlim`. The limiter is evaluated on each element from the element mean of n_{C0} and the element gradient, and the element flux $F_e = \langle|\nabla V|^2\rangle_e D_e\, \Delta n_{\mathrm{C0}}/\Delta V_e$ enters the two rows of the element as $+F_e$ and $-F_e$, so the term is exactly conservative. Evaluated at the nodes, $|\nabla n_{\mathrm{C0}}|/n_{\mathrm{C0}}$ is 0/0 at the separatrix, where the source and hence n_{C0} vanish; rounding then decided whether that node got the unlimited D_{C0}, and the bundle inventory at `NRMAX` = 120 came out 29% above 60 and 240. The limiter is part of the model: without it the known C^0 flux divergence, a source that does not scale with $\bar{n}_z$, drives $\bar{n}_z$ negative just outside the separatrix.

5.4 Larsen limiter on the hydrogen slow neutrals (optional; open issue)

`MDLNOFL` = 1 applies the same limiter to D_{01}. It is off by default in V5.60, which is **not** a statement that it is unnecessary. In the SOL the ratio $R = D|\nabla n|/(\alpha v n)$ is 10^3–2×10^3 (up to 2×10^4), so the plain diffusion flux far exceeds the free-streaming limit. With the limiter, the Picard map has a multiplier near $-R/(1+R) = -1$ and the iteration ends on a period-two cycle. Under-relaxing the three neutral densities alone (`RLXNO` = 0.5, used only with `MDLNOFL` = 1) restores convergence in most steps, but a few steps still reach `ICMAX` at $\varepsilon = 10^{-6}$. The effect of the limiter on the plasma is a few per cent (neutral inventory 25%, ionization source 1–3%, n_{e}(wall) 3%, Z_{eff} 0%). It is left for a better nonlinear treatment, another flux-limited diffusion, a moment model or Monte-Carlo neutral transport.

5.5 New defaults and V5.52 compatibility

parameter	V5.52	V5.60	meaning
`FSLP`	0.3	1.0	k_L
`MDLSOL`	(0)	1	ramp width δ
`MDLNZD`	(0)	1	bundle transport, M5
`MDLWBC`	(0)	1	wall density condition
`MDLNOFL`	(0)	0	Larsen limiter on D_{01}
`FLlim`	–	0.25	limiter factor α (C^0; D_{01} with `MDLNOFL` = 1)
`RLXNO`	–	0.5	neutral-only relaxation (`MDLNOFL` = 1 only)

These are all the defaults that differ. `parm/txparm.v552default` sets `FSLP` = 0.3, `MDLSOL` = `MDLNZD` = `MDLWBC` = `MDLNOFL` = 0. With it V5.60 is not bit-identical to V5.52.20 (the optimized build schedules the changed routines differently), but one step from the same state gives double-precision states that agree to 10^{-12} (median over the variables) and 10^{-9} at worst, except $\dot{\psi}_t$ at its Picard noise floor (3×10^{-7}); the profile output agrees to its single-precision resolution. The table `data/Carbon-S0F0_20260921.txt` is read at start-up in either case.

5.6 Other changes since V5.51

- `MDLPCK` = 2 (default): block-tridiagonal solver with one step of iterative refinement; the Picard floor falls from about 10^{-5} to 5×10^{-7} on the worst equation.
- `oldmixa` (default 0.5): Picard under-relaxation applied only on time steps whose sweep is measured not to contract.
- BDF2 keeps its history across `CONT`; one backward-Euler step is taken where there is none (cold start, `LOAD`).
- Analytic axis limit of the Pfirsch–Schlüter Fourier factors, $f_m(\cdot, 0) = 0$; $\hat{q}^2$ at the axis from the interior instead of $q(0)^2$.
- D_{0z} uses the impurity neutral speed V_{0z}.
- `FSTPTM` = 3 (option): multi-ion composition channel for the turbulent particle flux.
- Finite-collisionality closure for the higher-order flows; triangularity of grid-traced equilibria regularized; G-EQDSK p' normalization corrected; `ipbtdir` = 1 by default.
- Diagnostic switches, off by default: `iaxproj` (near-axis regularity projection), `Dnfac` (particle diffusion in the impurity continuity equation).